

## Ideation Phase

### Define the Problem Statements

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Date          | 01 October 2026                                                     |
| Team ID       | SWTID-2026-8331                                                     |
| Project Name  | Script-Controlled ACL – Restrict Record Access Based on Field Value |
| Maximum Marks | 2 Marks                                                             |

Customer Problem Statement:

Create a problem statement from the customer's point of view. For Script-Controlled ACL, focus on the security administrator or application owner who needs field-value-based control over record access.

A clear problem statement identifies the customer, the access-control goal, the barrier, its root cause, and the resulting impact. These points are used to define the problem before designing the ACL solution.

|                                                                                                                   |                   |                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I am</b><br><br><b>I'm trying to</b><br><br><b>but</b><br><br><b>because</b><br><br><b>which makes me feel</b> | Customer / role   | <b>Security administrator / application owner</b><br><small>The person responsible for protecting records and defining access rules.</small>                                               |
|                                                                                                                   | Goal / Job        | <b>Restrict records according to a field value</b><br><small>Allow authorized users to access records only when the configured condition is satisfied.</small>                             |
|                                                                                                                   | Barrier / problem | <b>Broad or static access rules may expose records unnecessarily</b><br><small>Role-only permissions may not capture the state or sensitivity stored inside each record.</small>           |
|                                                                                                                   | Root cause        | <b>Access decisions are not always evaluated against record data</b><br><small>The system needs a rule that can inspect a protected field before allowing the requested operation.</small> |
|                                                                                                                   | Emotional impact  | <b>Concerned about unauthorized or inconsistent access</b><br><small>Unclear access behaviour makes security management, auditing and user trust more difficult.</small>                   |

Project-Specific Problem Statement Example:

|                          |                                       |                                          |                                              |                                     |
|--------------------------|---------------------------------------|------------------------------------------|----------------------------------------------|-------------------------------------|
| <b>I am</b>              | <b>I'm trying to</b>                  | <b>But</b>                               | <b>Because</b>                               | <b>Which makes me feel</b>          |
| a security administrator | restrict record access by field value | static role-based rules can be too broad | access may depend on the actual record state | concerned about unauthorized access |

| Problem Statement (PS) | I am (Customer)                                   | I'm trying to                                                            | But                                                                         | Because                                                                                      | Which makes me feel                                                            |
|------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| PS-1                   | I am a security administrator / application owner | I'm trying to restrict access to records based on a specific field value | But broad or static role-based access can expose records that should remain | Because the access decision may not evaluate the actual value stored in the protected record | Which makes me feel concerned about unauthorized or inconsistent record access |

|      |                                                    |                                                                                       |                                                                                  |                                                                                                        |                                                                                            |
|------|----------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|      |                                                    |                                                                                       | restricted                                                                       |                                                                                                        |                                                                                            |
| PS-2 | I am a developer implementing secure record access | I'm trying to apply a reusable field-value condition before allowing a read or update | But manually repeating access checks can make rules harder to maintain and audit | Because the application needs a consistent mechanism to evaluate the record field during authorization | Which makes me feel uncertain about whether every protected record is handled consistently |